

Flash-a-card: An Interactive Flashcard Quiz Website for Student Review

Problem Statement:

Ever had that feeling where you need to review your notes but can't seem to understand anything or don't even know how to start? This often leads to lower test scores and stress. Our system will provide an interactive quiz using flashcards that shows questions, gives explanations after each answer, and helps students focus on the areas they get wrong.

Project Objectives:

- Provide an interactive quiz using flashcards.
- Show explanations for every question.
- Sort questions into correct and wrong piles to focus on review.
- Allow users to retry questions they got wrong.
- Help students improve knowledge efficiently through repeated practice.

Planned Features:

1. Card presentation
 - Front: prompt/question.
 - Back: short answer + **explanation** (shown immediately after the user submits an answer — regardless of correctness).
 - Optional media: image/audio/formula support on front or back.
2. Immediate feedback & explanations
 - User answers (multiple choice or true-false).
 - Immediately show: correct answer, short explanation, and a short tip
 - If the user was correct, option to rate confidence (easy/medium/hard).
 - Correct / Incorrect piles
 - Each card flows into either the **Correct** pile (mastered) or **Incorrect** pile (needs practice).
 - Cards in the Incorrect pile are available to retry during the session (user toggles retry now / later).
3. Retry workflow
 - Option after a session: "Retry incorrect cards" — runs a focused mini-session with only Incorrect pile cards.
 - After each retry, the card can be moved to Correct if answered correctly twice
4. Categorization & navigation
 - Browse by Subject or search across Topics.
 - Filter by difficulty, tags, or date added.
5. Shuffle & ordering
 - Shuffle the entire deck or session for variety.
6. Progress summary & analytics

- Session summary at end: total cards, correct, incorrect, accuracy %, time spent, avg time/card.
 - Per-topic streaks, long-term accuracy graphs, and a “Next review” suggestion (based on performance & confidence ratings).
7. Usability & UI elements
- Minimalist card view with large progress bar, counters for Correct/Incorrect, and prominent Next/Flip buttons.
 - Keyboard shortcuts: space to flip, 1/2/3 to answer/mode-select, R to retry incorrect.
8. Session persistence & resume
- Sessions auto-save; user can pause and resume later.
 - When resuming, show a short recap of what was completed and remaining incorrect cards.
9. Accessibility & mobile-first
- High-contrast theme, screen-reader labels, and large touch targets.
 - Responsive layout for phones and tablets

Planned Inputs and Outputs:

A. Inputs

- Select subject
- Select topic
- Answer a question (multiple choice)
- Mark question as complete
- Request next question
- Retry wrong questions
- Shuffle questions

B. Outputs

- Display list of topics for the selected subject
- Load quiz questions for selected topic
- Show correct/incorrect feedback + explanation
- Update correct/wrong pile
- Show next flashcard/question
- Display all questions answered incorrectly
- Show questions in random order

Logic Plan (Pseudocode)

Start Program

Display list of subjects

User selects a subject

Display list of topics within that subject

User selects a topic

Load quiz questions for the selected topic

Initialize empty piles: CorrectPile, WrongPile

Repeat for each question:

- Display question

- Wait for user answer

- If answer is correct:

 - Show explanation

 - Add question to CorrectPile

- Else:

 - Show explanation

 - Add question to WrongPile

- Ask if user wants next question

End Repeat

Display summary:

- Number of correct answers

- Number of wrong answers

- Option to retry WrongPile questions

End Program

References

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